Dok. 260 — Structural Comparison:
Universal Constraint and FFGFT

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May 2026

Abstract

The Universal Constraint framework (UC, D. Haskins) and the Fundamental Fractal Geometric Field Theory (FFGFT, J. Pascher) are systematically compared. Both frameworks derive $E \propto 1/r_{\text{min}}$ and $\hbar$ as an invariant action scale from structural principles — without presupposing spacetime. Numerical verifications of cosmological predictions, an internal consistency point in the UC density papers, and the question of justification for the UC starting point are discussed.

.1 Starting Points and Axiomatics

Universal Constraint (UC)

The UC framework begins from a universal admissibility condition

$$C_\alpha \Psi = 0, \tag{.1}$$

which defines the admissible state space $\mathcal{S}$. Coarse-graining produces equivalence classes $[\Psi]$ (macrostates), from which entropy, temporal ordering, and a minimal distinguishability scale r_{min} are derived. Closure of admissible transformations yields the invariant action scale $\hbar$ and cosmological quantities from a single infrared scale R_H.

The form of C_α remains unspecified in the existing UC papers. The framework describes consequences of an admissibility condition without deriving why that condition is the correct starting point.

FFGFT

FFGFT begins from a T^4 identification torus with a single free parameter $\xi = 4/30000$. The foundational relation

$$\tilde{T} \cdot m = 1 \tag{.2}$$

on T^4 is not postulated but motivated by the requirement that no fundamental contradiction can exist between QM and GR. T^4 is the minimal geometric structure that carries both descriptions. The value $\xi = 4/30000$ appears independently in three distinct contexts:

- Higgs formula: $\xi = \lambda_h^2 v^2/(16\pi^3 m_h^2)$
- Fine-structure constant: $\alpha = e^2\mu_0 c/(4\pi\hbar)$
- Koide relation: $\xi_{\text{Koide}} = 1.372 \times 10^{-4}$ (+2.9% above ξ)

The starting point has a motivation outside itself.

.2 Structural Parallels

Energy as Inverse Length

UC: From the density bound $\rho \sim 1/r_{\text{min}}^2$, the sector count $N \sim R/r_{\text{min}}$, and $E \sim \rho/N$, one formally obtains $E \sim 1/r_{\text{min}}$. The proportionality constant remains open.

FFGFT: From action $S = h$ and the uncertainty relation on T^4:

$$E_{\text{bit}}(L) = \frac{\hbar c}{L}. \tag{.3}$$

The proportionality constant $\hbar c$ is derivable from the T^4 geometry with ξ.

Result: Identical scaling relation from two independent derivations. UC proves existence; FFGFT supplies the proportionality constant.

$\hbar$ as a Normalisation Quantity

UC derives $\hbar$ as an invariant action scale from closure of admissible transformations: $E \cdot t =$ const under rescalings gives $E \cdot t = \hbar$.

FFGFT treats $\hbar$ explicitly as an SI conversion factor, not as an ontological primitive (Dok. 001a, Dok. 230).

Agreement: Both frameworks reject $\hbar$ as a fundamental law of nature. This is a non-trivial shared claim.

Numerical verification:

$$E \cdot t = \frac{\hbar c}{r} \cdot \frac{r}{c} = \hbar \tag{.4}$$

At $r = \ell_P$: $E = 1.956 \times 10^9$ J (Planck energy), $E \cdot t = 1.055 \times 10^{-34}$ J s $= \hbar$ ✓.

.3 Numerical Verification of UC Cosmology

UC derives cosmological quantities from a single infrared scale R_H. Table 1 shows numerical results for two choices of R_H.

Quantity	without α	with α	Observed	Diff.
H_0 [s^{-1}]	6.81×10^{-19}	2.20×10^{-18}	2.20×10^{-18}	$+0.2\%$
a_0 [m s^{-2}]	2.04×10^{-10}	6.61×10^{-10}	1.20×10^{-10}	$+450\%$
Λ [m^{-2}]	5.17×10^{-54}	5.41×10^{-53}	1.11×10^{-52}	-51%
m_{min} [kg]	8.0×10^{-70}	2.6×10^{-69}	—	—

Table 1: UC cosmology: starting from $R_H = R_{\text{part. horiz.}} = 4.40 \times 10^{26}$ m; $\alpha = R_p/R_{\text{Hubble}} = 3.24$. "With α": $H_0 \to \alpha c/R_p$, $\Lambda \to \alpha^2/R_p^2$.

H_0 agrees to 0.2% with the α-correction. Λ remains a factor of 2 too small (-51%). a_0 worsens with α: without α, a factor of 1.7 too small; with α, a factor of 5.5 too large. The α-correction from UC cosmo note 6 resolves the H_0 problem but introduces an a_0 problem — this remains open.

Comparison with FFGFT: $m_{\text{min}}(\text{UC}) \approx m_{\text{bit}}(\text{FFGFT}) \approx 10^{-70}$ kg (factor $3.2 = R_{\text{part. horiz.}}/R_{\text{Hubble}}$). Both frameworks give the same order of magnitude for the cosmic information energy.

.4 Internal Consistency Point in the UC Papers

Two different density scalings appear across the UC papers:

- `density_logic`: $\rho \sim 1/r_{\text{min}}^2$ (from the quadratic number of distinguishability relations $D \sim N^2$)
- `Scale_from_Geometry`: $\rho \sim 1/r_{\text{min}}^3$ (from volume $V(R) \sim R^3$ in $n = 3$)

These define two different ρ: a holographic surface density and a standard volume density. The derivation $E \sim \rho/N$ with $N \sim R/r_{\text{min}}$ gives:

$$\text{With } \rho \sim 1/r_{\text{min}}^2 : \quad E \sim \frac{1/r_{\text{min}}^2}{R/r_{\text{min}}} = \frac{1}{r_{\text{min}} \cdot R} \tag{.5}$$

The result $E \sim 1/r_{\text{min}}$ (independent of R) follows only if $R \sim r_{\text{min}}$ is assumed implicitly — an assumption not stated explicitly in the papers.

.5 Where UC and FFGFT Diverge

UC Stronger than FFGFT

UC derives from degeneracy consistency:

$$\text{Exclusion} \rightarrow \text{Antisymmetry} \rightarrow \text{Uncertainty}$$

The Pauli principle and $\Delta x \Delta p \geq \hbar/2$ follow from the requirement that probability $P_i = n_i/\sum n_j$ must correspond only to invariant structure. FFGFT takes quantum mechanics as given and does not derive it from T^4 geometry.

Also: $\Lambda \sim 1/R_H^2$ from constraint closure without vacuum energy density — structurally clean.

FFGFT Stronger than UC

- Concrete numerical predictions: all lepton masses from $m_i = r_i \cdot \xi^{p_i} \cdot v$ (numerically verified)
- Single free parameter $\xi = 4/30000$ — everything else follows
- Proportionality constants ($\hbar c$) derivable from T^4; UC leaves these open
- Non-Closure Theorem (Dok. 193): cross-products of different torus mode labels are structurally inadmissible — UC has no analogous precision

.6 Core Question for UC

UC begins with $C_\alpha \Psi = 0$. This condition is the foundation of all derived structure. A derivation of *why* this form of admissibility condition is the correct starting point — or what path leads to it — is not given explicitly in the existing papers.

FFGFT's starting point (T^4, ξ) has an external motivation: the independent appearance of ξ in three different contexts (Higgs, α, Koide) and the QM-GR compatibility requirement.

The question is not whether UC's results are correct — the structural parallels are real. The question is: *What path leads to $C_\alpha \Psi = 0$?*

.7 Summary

The result of the numerical verification and structural comparison suggests that UC and FFGFT do not stand in a hierarchical relationship. They overlap in central scaling claims but have different starting points, different strengths, and different open questions. Further collaboration — in particular on the justification of C_α and the derivation of proportionality constants in UC — appears worthwhile.

Property	UC	FFGFT
Starting point	$C_\alpha\Psi = 0$ (axiomatic)	$T^4 + \xi$ (motivated)
$E \propto 1/r_{\text{min}}$	✓(qualitative)	✓(with $\hbar c$)
$\hbar$ not fundamental	✓	✓
H_0 correct	✓($\pm 0.1\%$)	partial
Λ correct	factor 1.85 too small	open
Pauli exclusion derived	✓	$\times$
Numerical mass predictions	$\times$	✓
Free parameters	$\hbar, c, R_H$	ξ
Proportionality constants	open	from T^4

Table 2: Comparison overview: UC vs. FFGFT.